ACM Multimedia 2003 Workshop Proposal

Title: The 1st International Workshop on Video Surveillance

Organizers: Edward Chang and Yuan-Fang Wang
University of California, Santa Barbara

Motivation and Goals

The heightened concerns about national and personal security have intensified the search for effective, efficient surveillance. Being able to automatically discover and characterize suspicious events accurately and quickly will benefit both government agencies and private enterprises in combating terrorism and crimes.

With the proliferation of inexpensive cameras and the deployment of high-speed, wired/wireless networks, it has become economically and technically feasible to employ multiple cameras for security surveillance. In a surveillance system, video signals are generated by multiple cameras with or without spatially and temporally overlapping coverage. These signals need to be transmitted, processed, fused, stored, indexed, and then summarized as semantic events to allow efficient and effective queries and mining by security personnel and law enforcement officers.

This workshop will bring together researchers, developers and practitioners from academia and industry to discuss various issues involved in developing a large-scale video surveillance system: e.g., from low-level feature extraction to high-level object and motion description, from camera control to networks to storage and indexing, and from theory to practice.

Interest within the Multimedia Community

A primary mission for members of the Multimedia community is to increase our impact and influence. We believe that video surveillance research can contribute to this mission if the research leads to the development of "killer" applications of scholarly significance. If the Multimedia community does not take the initiative to *own* this exciting and challenging research area, other communities, such as Computer Vision and Signal Processing, will take the lead.

Building a large-scale video surveillance system requires a synergistic effort of multiple disciplines including, but not limited to, Computer Vision, Signal Processing, Machine Learning, Database Systems, and Network Management. There is no other research community better suited than the Multimedia group to lead this interdisciplinary research. Within the Multimedia community this research endeavor will foster close collaboration among its three current research tracks: Content, Systems, and Tools. We also believe that this research endeavor can draw new members to the ACM Multimedia community.

Program Committee Members (TBI)

- J. K. Aggarwal, University of Texas
- Shih-Fu Chang, Columbia University
- Navenka Dimitrova, Philips Research
- Yihong Gong, NEC Research
- Ajay Divakaran, Mitsubishi Research
- Ioannis Kakadiaris, University of Houston
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- Thomas Plagemann, Univ of Oslo, Norway
- Stan Sclaroff, Boston University
- Cyrus Shahabi, USC
- Prashant Shenoy, Univ of Massachusetts, Amherst
- John Smith, IBM T.J. Watson Research
- Mohan Trivedi, University of California, San Diego
- Hong-Jiang Zhang, Microsoft Research, China

Call for Paper — Sample

With the proliferation of inexpensive cameras and the deployment of high-speed, wired/wireless networks, it has become economically and technically feasible to employ multiple cameras for security surveillance. In a surveillance system, video signals are generated by multiple cameras with or without spatially and temporally overlapping coverage. These signals need to be transmitted, processed, fused, stored, indexed, and then summarized as semantic events to allow efficient and effective queries and mining by security personnel and law enforcement officers. This workshop will bring together researchers, developers and practitioners from academia and industry to discuss various issues involved in developing a large-scale video surveillance system: e.g., from low-level feature extraction to high-level object and motion description, from camera control to networks to storage and indexing, and from theory to practice.

The workshop will cover a variety of research issues on video surveillance. It will focus on the underlying theory, methods, systems, and applications. The topics will include, but are not limited to, the following:

- Video Content Analysis
  - Multi-Camera Calibration
  - Motion Detection and Tracking
  - Face Detection and Recognition in Unconstrained Environments
  - Object Recognition
  - Multi-camera/-sensor Fusion
- Video Event Modeling and Mining
  - 2D/3D Multi-level Scene/Pattern Representations
  - Machine Learning Techniques for Event Mining
  - Spatio-temporal Data Mining
  - Threat Assessment
- Video Indexing and Storage
  - Spatio-temporal Data Indexing
  - Archival & Retrieval of Video Data
  - Query Paradigms and Languages
  - Mobile Architectures
- Surveillance Applications
  - Military Applications
  - Civilian Applications

Proposed Format: The workshop will open with a special session consisting of three to four papers that will introduce and overview the research area. Presentations will then be organized into several sessions corresponding roughly to the categories identified above. The workshop will conclude with a round-table discussion on future research directions.

Proposed Length: One Day during which each session will be allocated 1 to 2 hours, depending on the number of contributions and the expected length of the discussion session.

Workshop Publications: The accepted papers will be available electronically from the workshop website, and also as printed workshop notes (published by ACM Multimedia).

A special issue of ACM Multimedia journal or IEEE Transaction on Multimedia (under discussion) will publish selected papers.

Submissions: Authors are invited to submit papers on the topics outlined above or on other related issues. Submissions should not exceed 8 pages, and should conform to the ACM style sheet. Electronic submissions, in PDF format, are required and should be sent to Prof. Edward Chang at echang@ece.ucsb.edu.

Important Dates:

- Submission Deadline: July 8, 2003
- Acceptance Notification: August 18, 2003
- Final date for camera-ready copies to organizers: August 31, 2003

Organizers

Prof. Edward Chang received his Ph.D. in Electrical Engineering at Stanford University in 1999. He joined the University of California at Santa Barbara as an Assistant Professor in 1999, and he has been a tenured Associate Professor of Electrical & Computer Engineering and of Computer Science since 2003. His current research interests include Statistical Learning and Multimedia Databases. Prof. Chang's work on Perception-Based Image Retrieval (PBIR), which applies statistical learning to formulate users' subjective similarity metrics, was recognized as a major breakthrough in image retrieval by the CBIR panel at the 2002 IEEE International Conference on Multimedia. He is a recipient of the IBM Faculty Partnership Award from 2000 to 2002, and the NSF Career Award in 2002. He is also a co-founder and the CTO of VIMA Technologies.

Professor Yuan-Fang Wang received his graduate training at the University of Texas at Austin. He has been with the Department of Computer Science at the University of California at Santa Barbara since 1987, where he is currently a full Professor. Professor Wang is very active professionally. He was the program co-chair of the 1998 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, an Associate Editor for IEEE Transactions on Pattern Analysis and Machine Intelligence from 1998 to 2002, and an Associate Editor for the Pattern Recognition Journal since 2000. Professor Wang also served as a consultant for a number of companies and was a faculty consultant at LG Electronics Research Center of America, Princeton Junction, New Jersey in summer 1998. He has published over 70 peer-reviewed papers in the area of digital libraries, computer vision, computer graphics, medical image analysis, and robotics.